

One neuron is a weighted sum followed by an activation

each input carries
its own weight

setpoint deviation

occupant density

months since service

Weighted sum

$$w_1x_1 + w_2x_2 + w_3x_3 + b$$

Activation

tanh, ReLU, sigmoid

Output

to the next layer

Without the activation the whole network collapses into a single weighted sum,
so a stack of linear layers can express nothing that one linear layer cannot.